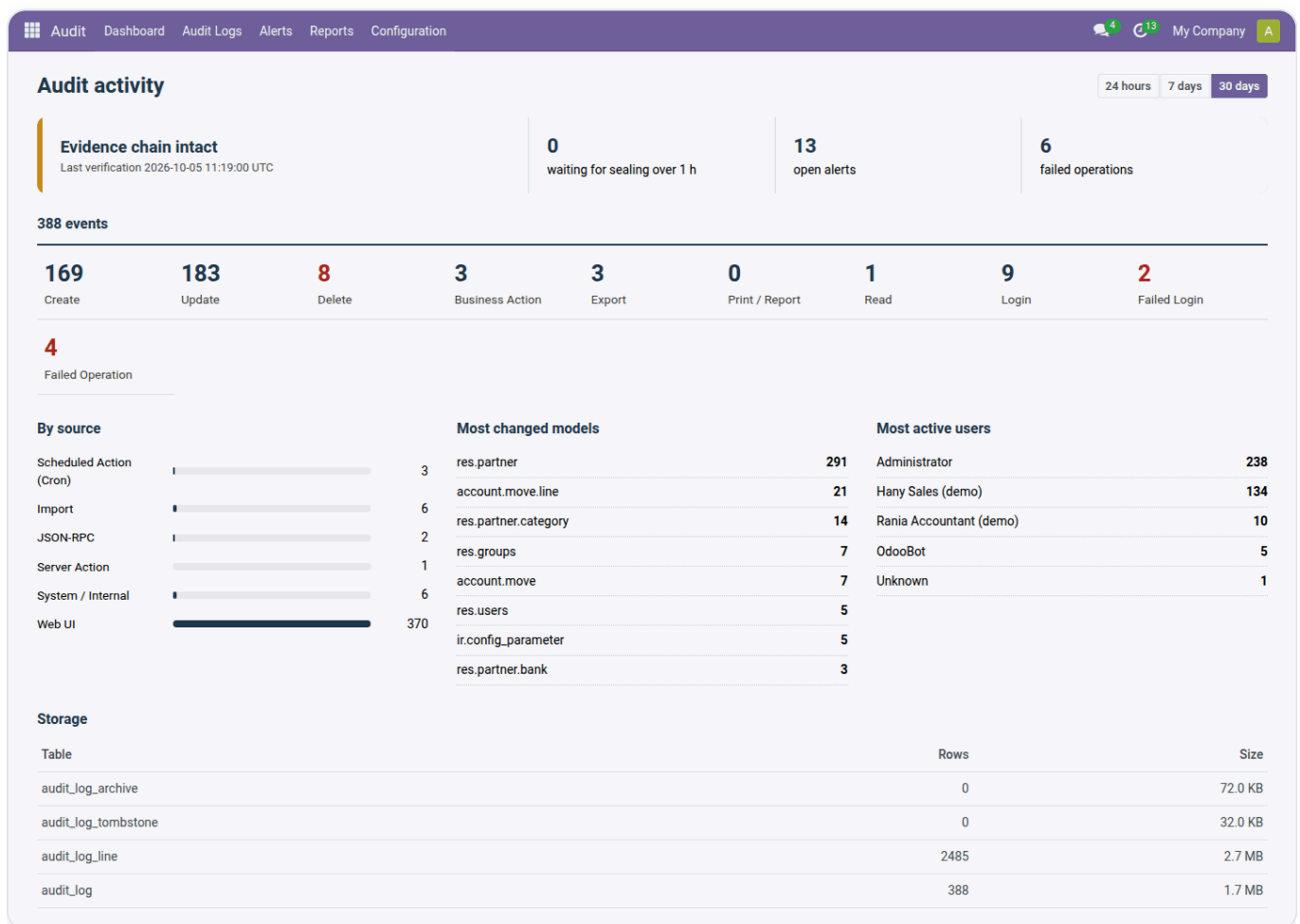


Audit Log & Compliance Engine

A tamper-evident audit trail for any Odoo model, with no per-model code: field-level changes, deleted-record forensics, business actions, data access, logins, alerts, hash-chain integrity, retention and governed redaction.



Module: `nx_audit_log` · Version `18.0.1.0.0` · Author: **Sayed Anwar** · October 2026

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This guide was written on Odoo 18 with the demo data module **nx_audit_log_demo** installed. Every screenshot is from a real database. Buttons look like this: ; menus look like this: **Audit → Configuration → Audit Rules**.

1. The problem this module solves

Odoo's chatter tracks a few fields on a few models. It does not tell you who deleted an invoice, who exported the full customer list, which API client read 850 contacts at 2 a.m., or whether someone changed a bank account number and changed it back. And chatter messages can be deleted by anyone with enough rights, so they are a convenience, not evidence.

Typical question from an auditor or manager	Answer with standard Odoo	Answer with nx_audit_log
Who changed this invoice total, from what to what?	Only if the field is tracked	Every field, old → new, user, IP, source, time
What did the deleted record contain?	Gone	Full delete snapshot + database cascade effects
Who exported or bulk-read our customers?	Not recorded	Export, print and bulk API read events
Who gave a user Administration rights?	Not recorded	Group change event + critical alert
Can we prove the log was not edited?	No	HMAC hash chain, daily verification
How long do we keep it, and can we erase personal data?	No policy	Retention policies + four-eyes redaction

The idea

The module records **evidence**, not messages. One event (**audit.log**) answers *who, when, where from and how*; its lines (**audit.log.line**) answer *which field, old value, new value*. Evidence is written in the same database transaction as the business change, so a change that is rolled back never appears, and a change that is committed always does. Nobody can edit or delete evidence from the interface, and a background job seals it into a hash chain that reveals any change made directly in the database.

What is captured

- **Changes:** create, update, delete, copy, with direct, derived (recomputed) and database-cascade origins; one2many lines and many2many links.
- **Business actions:** button methods such as *Confirm, Post, Archive*, linked to the field changes they caused.
- **Data access:** exports, printed reports, imports, and API reads above a threshold.
- **Authentication:** login, logout, failed logins (grouped per login, IP and time window), second factor.
- **Failures:** operations the user saw refused (e.g. deleting a posted entry).

- **Configuration:** every change to audit rules, alert rules, retention, sensitive patterns and audit settings.

2. Roles and access

Assign roles in **Settings → Users & Companies → Users**, section **Audit & Compliance**.

Roles and access

Each role includes the one below it. Side groups widen the view without giving more power.

Audit Administrator
Settings · audit rules · alert rules · sensitive fields · retention · redaction requests · maintenance buttons

Audit Manager
Alerts · reports · before/after snapshots · archived events · integrity checks and hash chains

Audit User
Dashboard and every audit log menu (events, field changes, batches)

See shared records
Events on records without a company (included in Audit User)

Cross-company auditor
Events of all companies, not only the allowed ones

Restricted-fields override
Lines of fields protected by groups= (e.g. HR, payroll)

⚠ Nobody, not even the administrator, can create, edit or delete evidence from the interface.

Each role includes the one below it

Group	Typical user	Can do
Audit User	Team leader, internal control	Dashboard, every Audit Logs menu, record timelines
Audit Manager	Compliance officer, internal auditor	+ Alerts, Reports, snapshots, archived events, integrity checks
Audit Administrator	IT / system owner	+ Configuration: settings, audit rules, alert rules, sensitive fields, retention, redaction
Audit: see shared / global records	Implied by Audit User	Events on records without a company
Audit: cross-company auditor	Group auditor	Events of all companies
Audit: compliance override for restricted fields	HR / payroll auditor	Lines of fields restricted with groups=

Company visibility: an event is visible in every company it touched (the record's company before and after the change). **Field visibility:** a line keeps the groups of its source field, so a user who cannot see salary in HR cannot see it in the audit log either.

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3. Installation and first setup

- 1 **Install** `nx_audit_log` from **Apps**. It depends only on `base`, `web`, `mail` and `base_import`.
- 2 **Set the integrity key** in the server configuration file and restart Odoo:


```
nx_audit_hmac_key = <64 random hex characters> # e.g. openssl rand -hex 32
```

Without it, the module generates a key in the database and the settings show a warning.
Keep the key with your backup procedure: without it, a restored database cannot be verified.
- 3 **Review the default rules** in **Audit → Configuration → Audit Rules**. Rules are created for the apps installed at that time. After installing a new app (Accounting, Sales, Inventory...) click **Load default rules** in **Audit → Configuration → Settings**.
- 4 **Decide on the broad rule.** *All business models (Create / Update / Delete)* ships **inactive**. Check its excluded models, then activate it if you want near-total coverage. Explicit model rules always win over it.
- 5 **Give roles** (chapter 2) and set the **recipients** of the alert rules (chapter 8).
- 6 **Optional:** install `nx_audit_log_demo` on a test database to explore realistic evidence. Uninstalling it removes every demo record and demo event.

Audit Dashboard Audit Logs Alerts Reports Configuration										
New Audit Rules Search...										
☐	Name	Scope	Model	Create	Update	Delete	Business Ac...	Field Mode	Retention Policy	Active
☐	Journal entries & invoices	One model	Journal Entry	☒	☒	☒	☒	All fields	Finance (10 years, no purge)	☒
☐	Bank accounts	One model	Bank Accounts	☒	☒	☒	☒	All fields	Finance (10 years, no purge)	☒
☐	Contacts	One model	Contact	☒	☒	☒	☒	All fields	Standard (3 years)	☒
☐	Users & access	One model	User	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	Security groups	One model	Access Groups	☒	☒	☒	☒	Selected fields only	Security (5 years)	☒
☐	Access rights	One model	Model Access	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	Record rules	One model	Record Rule	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	Companies	One model	Companies	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	API keys	One model	Users API Keys	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	System parameters	One model	System Parameter	☒	☒	☒	☒	All fields	Security (5 years)	☒
☐	Attachments (metadata only)	One model	Attachment	☒	☐	☒	☒	Selected fields only	Standard (3 years)	☒
☐	Demo: contact tags	One model	Partner Tags	☒	☒	☒	☒	All fields		☒
☐	Payments	One model	Payments	☒	☒	☒	☒	All fields	Finance (10 years, no purge)	☒
☐	Chart of accounts	One model	Account	☒	☒	☒	☒	All fields	Finance (10 years, no purge)	☒
☐	Sales orders	One model	Sales Order	☒	☒	☒	☒	All fields	Standard (3 years)	☒
☐	All business models (Create / Update / Delete)	All eligible models		☒	☒	☒	☐	All fields		☐

Default audit rules with their retention policy

Default rules

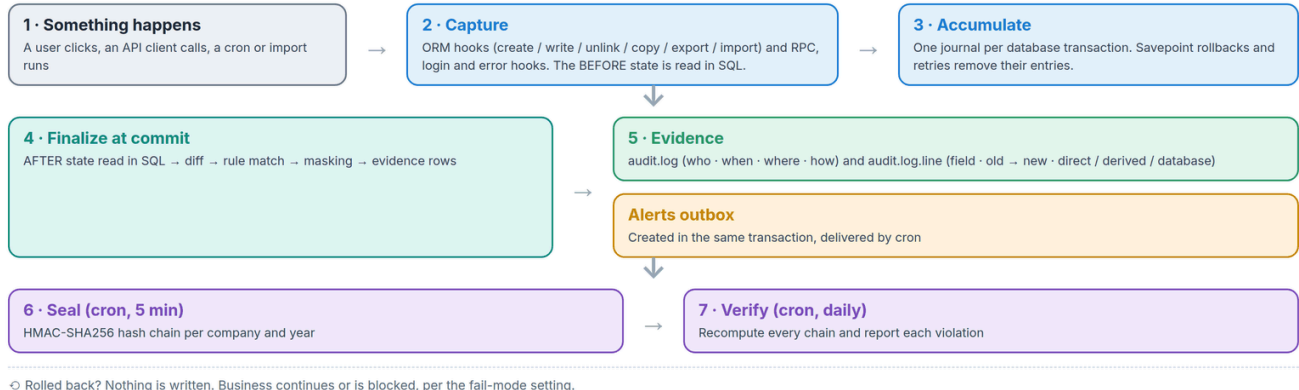
Rule	Model	Special settings	Retention
------	-------	------------------	-----------

Journal entries & invoices	account.move	Print/report on, full snapshot	Finance
Payments · Chart of accounts · Bank accounts	account.payment · account.account · res.partner.bank	Payments: print/report on	Finance
Sales orders · Purchase orders · Transfers	sale.order · purchase.order · stock.picking	Orders: print/report on	Standard
Stock quantities	stock.quant	Selected fields: quantity, location, lot, package, owner	Standard
Contacts	res.partner	Direct changes only	Standard
Users & access · Security groups · Access rights · Record rules · Companies	res.users · res.groups · ir.model.access · ir.rule · res.company	Password is secret	Security
API keys · System parameters	res.users.apikeys · ir.config_parameter	Key value is secret	Security
Attachments (metadata only)	ir.attachment	Name, type, size, linked record; no content	Standard

4. How capture works

How every change becomes evidence

No per-model code: the engine sits under the ORM and at the request boundary, and writes evidence in the same transaction as the business change.



From a click to sealed evidence

- **Before** values are read with SQL at the first touch of a record, and **after** values with SQL after the final flush. Evidence is what is really stored, not what the screen showed. Writing a field back to its original value produces no change.
- **Change origin** on every line: **Direct** the caller wrote the field; **Derived** Odoo recomputed it (e.g. invoice total after a price change); **Database cascade** PostgreSQL changed it (e.g. a contact's parent set to empty when the company is deleted).
- **Child lines**: a rule on a model also audits the lines it owns (one2many with cascade delete). The *Journal entries* rule audits invoice lines; the *Sales orders* rule audits order lines.
- **Source** is decided by the server, not by the client: Web UI, JSON-RPC, XML-RPC, external controller, cron, import, automated action, server action, system. A client cannot send a context key that disables auditing or fakes its source.
- **Savepoints and retries**: work undone by a savepoint (including an import test with *Test*) or a retried transaction never becomes evidence.
- **Bulk operations**: one write or delete touching at least *Bulk operation threshold* records is grouped into a **batch** with counters.

5. Settings in detail

Audit → Configuration → Settings (Audit Administrator). Every change of these settings is itself recorded as a **CONFIG** event.

Audit

Dashboard

Audit Logs

Alerts

Reports

Configuration

SaveDiscardSettings

Search...

General Settings

Sales

Audit

Invoicing

Text previews

Characters kept in readable previews; the structured value is the evidence.

300

Evidence language

Language used for record names stored in evidence.

en_US

If auditing fails

Keep business running and record an engine error

Block the transaction

Large values

Longer text/HTML is stored as a digest plus its beginning.

20,000

Excluded fields

Technical fields never audited unless a rule selects them explicitly.

Audit every export

Bulk API read threshold

API reads returning at least this many records are logged even without a read rule.

500

Bulk operation threshold

One write/delete touching this many records is grouped into a batch.

100

Failed login window

Failed logins are counted per login, IP and window.

10

Store unknown login names

Sealing delay

Seconds before an event is sealed into its hash chain.

120

Integrity key

Maintenance

The Audit block of the settings

Setting	Default	What it does / recommendation
Text previews	300	Characters kept in the readable <i>Old / New value</i> columns. The full structured value is stored separately and is the real evidence.
Large values (max stored text length)	20 000	Longer text or HTML is stored as a digest + its beginning, to keep tables small. Raise it only if you must keep long texts in full.
Evidence language	en_US	Language used for record names and selection labels saved in evidence, so evidence does not depend on each user's language.

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Excluded fields	write_date, write_uid, create_date, create_uid, message_ids, activity_ids, parent_path, standard_price, access_token...	Technical fields never audited, unless a rule selects them explicitly in <i>Selected fields only</i> .
If auditing fails	Keep business running	Keep business running and record an engine error: availability first; an ENGINE_ERROR event is written separately. Block the transaction: compliance first; the user's save is refused if evidence cannot be written.
Audit every export	On	Every <i>Export</i> is recorded even on models without a rule (fields exported and number of records).
Bulk API read threshold	500	An API read returning at least this many records is recorded as a bulk read, even without a read rule. Detects data scraping.
Bulk operation threshold	100	A single write/delete touching this many records is grouped into a batch.
Failed login window	10 min	Failed logins are counted per login + IP + window, so 1 000 attempts make one event with an attempt count, not 1 000 rows.
Store unknown login names	Off	Off: a typed login matching no user is stored hashed (people sometimes type their password in the login field). Turn on only if your policy allows it.
Sealing delay	120 s	Time before an event is sealed into its chain, so late-committing transactions are sealed in order.
Integrity key	-	Shows where the key comes from: <i>Server configuration file</i> (recommended) or <i>Database</i> (warning).
Maintenance buttons	-	Load default rules after installing apps · Seal pending events now instead of waiting for the cron · Enable value search indexes (PostgreSQL pg_trgm) for fast search in old/new values on large databases.

6. Audit rules in detail

Audit → Configuration → Audit Rules. A rule says *what* to record on *which* model, for *whom*, and how long to keep it. When several rules match, the lowest **Sequence** wins. The smart button **Events** opens what the rule recorded.

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4

13

My Company

A

New

Audit Rules

Journal entries & invoices

Events

28

1 / 1

<

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Name

Journal entries & invoices

Scope

One model

All eligible models

Model

Journal Entry

Sequence ?

5

Retention Policy

Finance (10 years, no purge)

Snapshot Policy ?

Full before/after snapshot

Chatter Mirror ?

☐

Operations

Fields

Scope

Notes

CHANGES

Create

Update

Delete

Business Actions

Action Methods ?

☒

☒

☒

☒

action_post, button_cancel

DATA ACCESS

Export

Print / Report

Read ?

☒

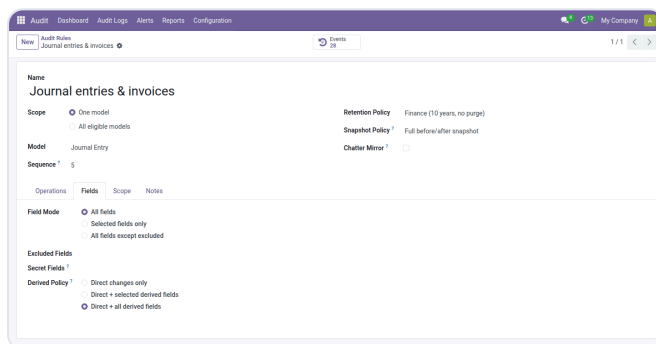
☒

☐

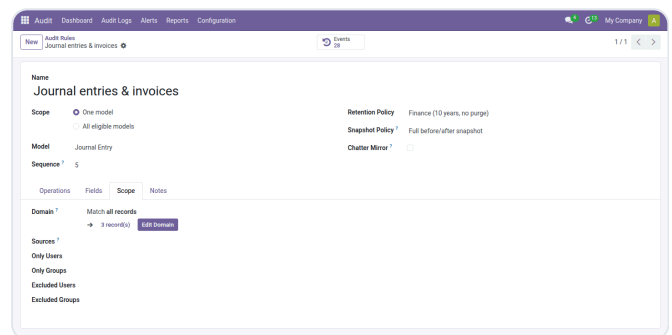
Rule form, Operations tab

Field	Meaning
Scope	<i>One model</i> , or <i>All eligible models</i> (with Excluded Models). Technical, transient and infrastructure models can never be audited.
Model · Sequence · Active	The audited model; priority between rules; switch the rule off without deleting it.
Retention Policy	How long the events of this rule stay online and in total (chapter 11).
Snapshot Policy	<i>Changed fields</i> , or <i>Full before/after snapshot</i> of the record at each change. Deleted records always keep a delete snapshot.
Chatter Mirror	Also post a short summary on the record chatter. The audit tables stay the evidence.
Operations tab	Create , Update , Delete , Business Actions , Export , Print / Report , Read . <i>Read</i> is aggregated per request and expensive: use it only for very sensitive models. Action Methods : comma-separated method names (e.g. <code>action_post, button_cancel</code>); empty = every button method.
Fields tab · Field Mode	<i>All fields</i> , <i>Selected fields only</i> (Audited Fields), or <i>All fields except excluded</i> (Excluded Fields).

Secret Fields	The change is recorded, the value is never stored (shown as masked).
Derived Policy	<i>Direct changes only</i> · <i>Direct + selected derived fields</i> (Derived Fields) · <i>Direct + all derived fields</i> . Example: on invoices keep <i>all</i> so a changed price also shows the recomputed total; on contacts <i>direct</i> avoids noise.
Scope tab · Domain	Only records matching the domain before <i>or</i> after the change (stored fields only). Example: only customer invoices <code>[('move_type', '=', 'out_invoice')]</code> .
Sources	Record only from some sources (e.g. only JSON-RPC to watch an integration). Empty = all.
Only Users / Only Groups · Excluded Users / Groups	Limit the rule to, or exclude, some people (e.g. exclude an integration user that is audited elsewhere).
Companies	Limit the rule to some companies. Empty = all.



Fields tab: field mode, secret fields, derived policy



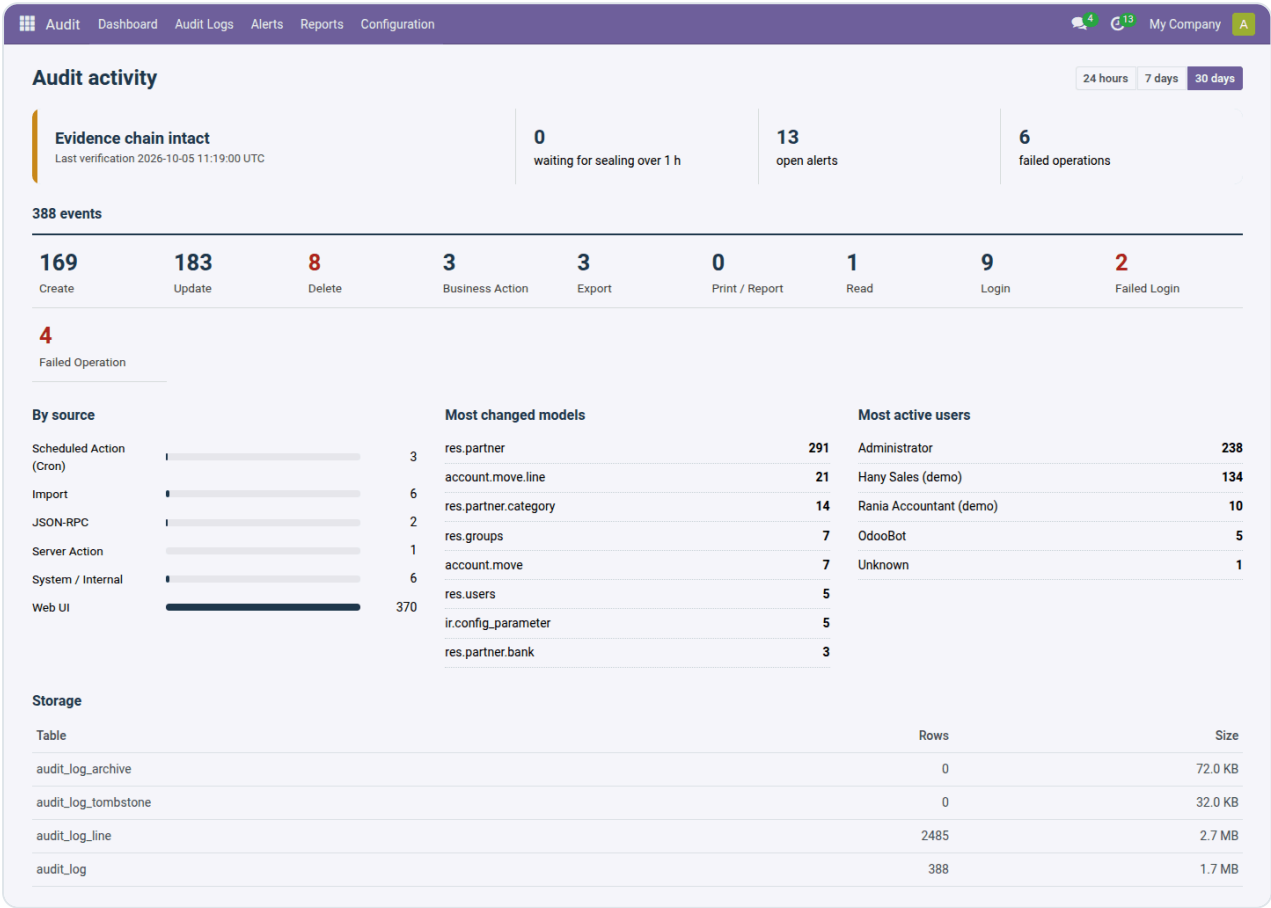
Scope tab: domain, sources, users, groups

Quick access: every audited model gets an **Audit Trail** entry in the **Action** ⚙️ menu of its records, which opens the record timeline.

7. Reading the evidence

7.1 Dashboard

Audit → Dashboard: period 24 hours / 7 days / 30 days; integrity status (click to open the last verification); unsealed backlog; open alerts; failed operations; counts per operation; events by source; most changed models; most active users; table sizes. Every number opens the matching events.



Dashboard over 30 days

7.2 Audit Logs menus

Audit → Audit Logs: All Events, Creates, Updates, Deletes, Business Actions, Data Access, Authentication, Failed Operations, Batches, Archived Events. Each list can be filtered and grouped by model, user, source, IP, company, date. Deletions and failures are shown in red.

Audit Dashboard Audit Logs Alerts Reports Configuration 4 13 My Company A									
Audit Logs Q Last 7 days × Search... ▼ 1-80 / 337 < > ☰ ☲ ☱									
Reference	Event Datetime	Operation	Model	Record	Action	Fields	User	Source	
☐ AUD/2026/00000003...	10/05/2026 14:19:43	Update	Companies	My Company		1	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:19:38	Update	User	Administrator		1	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:19:37	Login	User	admin	mfa: default	0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:19:37	Update	Contact	Administrator		1	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Export	Contact			0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Failed Operation	Bank Accounts	EG380019000500000000263180002		0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Bank Accounts	EG380019000500000000263180002		6	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Update	Sales Order Line	S00002 - Industrial Pump IP-200 (Nile Pharma)		4	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Update	Sales Order	S00002		3	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Sales Order Line	S00002 - Industrial Pump IP-200 (Nile Pharma)		21	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Sales Order	S00002		19	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Delete	Journal Entry	Draft Invoice		0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Delete	Journal Item	Draft Entry		0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Delete	Journal Item	15%		0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Delete	Journal Item	Filter Cartridge FC-10		0	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Journal Item	Draft Entry		17	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Journal Item	15%		20	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Journal Item	Filter Cartridge FC-10		23	Administrator	Web UI	
☐ AUD/2026/00000003...	10/05/2026 14:18:39	Create	Journal Entry	Draft Invoice		28	Administrator	Web UI	

All events, last 7 days

7.3 The event form

Header: reference (e.g. AUD/2026/0000000352), operation, record, seal state, and buttons **Record timeline** and **Open record** . **What:** model, record ID, rule, change origin. **Who / when:** date, user, login, initiating user (when someone acts for another), superuser mode, companies, result.

Tab	Content
Changes	One line per field: change type (set, change, clear, added, removed, line created/updated/deleted), old value, new value, origin.
Resulting changes	For a business action: the field changes it caused.
Child events	Events on owned lines or records changed by a database cascade.
Snapshot	Before / after / delete snapshot (Audit Manager).
Access details · Request	Fields and record IDs read or exported; source, execution path, IP address, user agent, claimed client, request path, session hash, correlation ID.
Integrity & retention	Chain, sequence, hash, previous hash, key ID; retention policy, archive and purge dates, redaction flag.

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My Company

A

Audit Logs

AUD/2026/000000043 ACTION Red Sea Logistics

Same request

1 / 1

<

>

Record timeline

Open record

Unsealed

Sealed

AUD/2026/000000043

Business Action

Red Sea Logistics

Archive

WHAT

Model

Contact

Model

res.partner

Record ID

19

Record Count

1

Rule

Contacts

Change Origin

Semantic

WHO / WHEN

Event Datetime

09/26/2026 04:16:55

User

Hany Sales (demo)

User Login

demo.sales@nextera.demo

Superuser Mode

☐

Company

My Company

Visible to Companies

My Company

Result

Success

Changes

Resulting changes

Snapshot

Access details

Request

Integrity & retention

Reference	Event Datetime	Operation	Model	Record	Action	Fields	User	Source	
AUD/2026/000000044	09/26/2026 04:16:55	Update	Contact	Red Sea Logistics	Archive	1	Hany Sales (demo)	Web UI	

A business action (Archive) with its resulting change

Audit

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13

My Company

A

Audit Logs

AUD/2026/0000000308 READ res.partner

Same request

1 / 1

<

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Unsealed

Sealed

AUD/2026/0000000308

Read

WHAT

Model

Contact

Model

res.partner

Record ID

0

Record Count

850

Rule

Change Origin

WHO / WHEN

Event Datetime

10/01/2026 08:02:55

User

Rania Accountant (demo)

User Login

demo.accountant@nextera.demo

Superuser Mode

☐

Company

My Company

Visible to Companies

My Company

Result

Success

Changes

Snapshot

Access details

Request

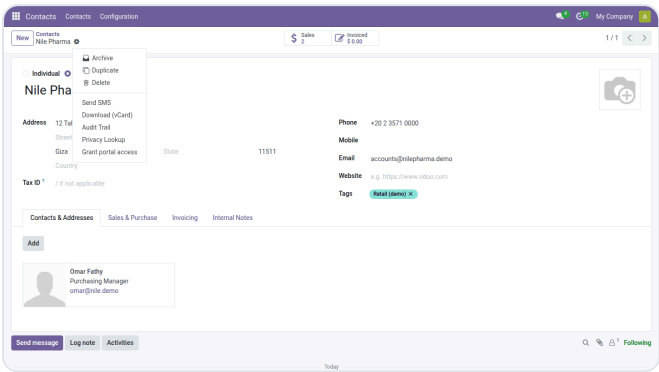
Integrity & retention

Source	JSON-RPC	IP Address	185.220.101.4
Execution Path	demo.nx_audit_demo	User Agent	python-requests/2.32
Request Path	/jsonrpc	Claimed Client ?	
Http Method	POST	Session Hash	c830bc09951a4013908bb57c
Correlation	53a7297ed70142d38a2d09c3c779026e		

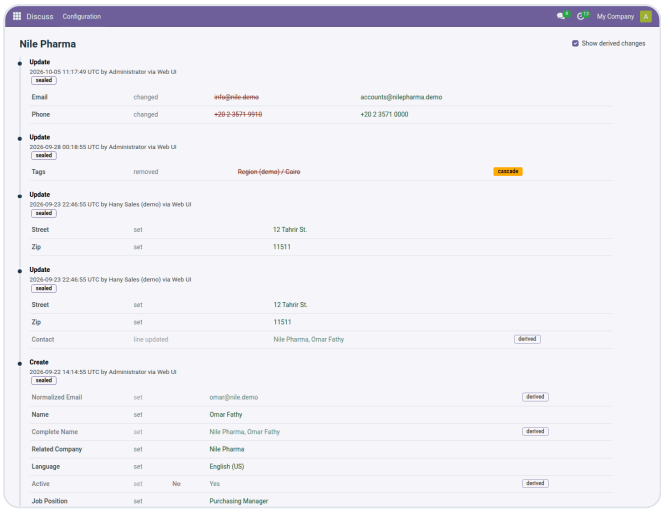
A bulk API read: 850 contacts read by python-requests over JSON-RPC

7.4 Record timeline

From an event click **Record timeline** , or from any audited record use **Action** ⚙️ → **Audit Trail**. The timeline shows every event on the record and on its lines, newest first, with old values struck through. Tick **Show derived changes** to also see recomputed fields.



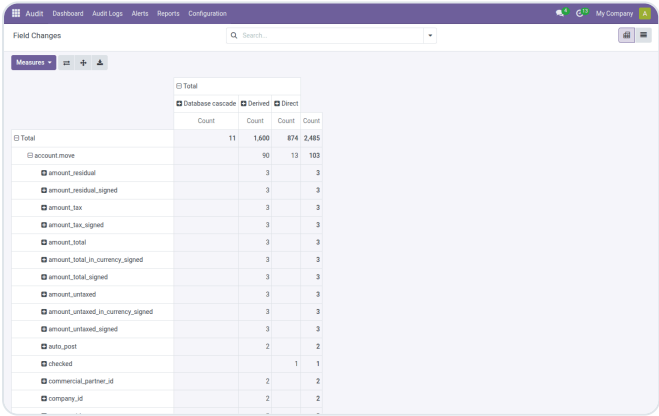
Audit Trail in the Action menu of a contact



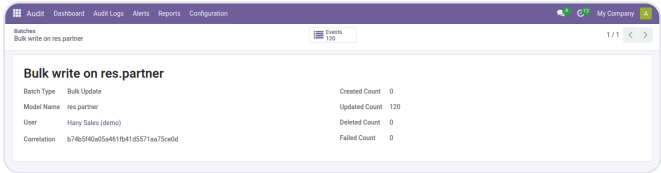
The record timeline

7.5 Field changes and batches

Audit → **Reports** → **Field Changes** analyses lines per model, field and origin (pivot / graph). **Audit** → **Audit Logs** → **Batches** groups bulk operations and imports with created / updated / deleted / failed counters.



Field changes pivot



A batch of 120 updates

Audit Dashboard Audit Logs Alerts Reports Configuration									
Authentication									
Reference	Event Date/Time	Operation	Model	Record	Action	Fields	User	Source	
AUD/2026/0000000385	10/05/2026 14:19:37	Login	User	admin	mfa: default	0	Administrator	Web UI	
AUD/2026/0000000347	10/05/2026 14:18:37	Login	User	admin	mfa: default	0	Administrator	Web UI	
AUD/2026/0000000337	10/05/2026 14:17:40	Login	User	admin	mfa: default	0	Administrator	Web UI	
AUD/2026/0000000336	10/05/2026 14:17:40	Login	User	admin	mfa: default	0	Administrator	Web UI	
AUD/2026/0000000335	10/05/2026 14:17:33	Login	User	admin	mfa: default	0	Administrator	Web UI	
AUD/2026/0000000330	10/05/2026 06:08:55	Logout	User	demo.accountant@nexusara.demo		0	Rania Accountant (demo)	Web UI	
AUD/2026/0000000329	10/05/2026 04:03:55	Failed Login	User	sha256:bf844866365981846659099		0		Web UI	
AUD/2026/0000000328	10/05/2026 02:05:55	Failed Login	User	admin		0	Administrator	Web UI	
AUD/2026/0000000327	10/05/2026 00:15:55	Login	User	demo.sales@nexusara.demo	mfa: default	0	Harry Sales (demo)	Web UI	
AUD/2026/0000000326	10/04/2026 22:07:55	Login	User	demo.accountant@nexusara.demo	mfa: default	0	Rania Accountant (demo)	Web UI	
AUD/2026/0000000325	10/04/2026 20:06:55	Login	User	demo.sales@nexusara.demo	mfa: default	0	Harry Sales (demo)	Web UI	

Authentication events

Audit Dashboard Audit Logs Alerts Reports Configuration									
Failed Operations									
Reference	Event Date/Time	Operation	Model	Record	Action	Fields	User	Source	
AUD/2026/0000000382	10/05/2026 14:18:39	Failed Operation	Bank Accounts	EG380010005000000000263180002		0	Administrator	Web UI	
AUD/2026/0000000368	10/05/2026 14:18:38	Failed Operation	Journal Entry	NNV/2026/00003		0	Administrator	Web UI	
AUD/2026/0000000332	10/05/2026 09:52:55	Failed Operation	Contact	Riyadh Medical Supplies		0	Rania Accountant (demo)	Web UI	
AUD/2026/0000000331	10/05/2026 07:51:55	Failed Operation	Contact	Al Nour Trading		0	Harry Sales (demo)	Web UI	
AUD/2026/0000000329	10/05/2026 04:03:55	Failed Login	User	sha256:bf844866365981846659099		0		Web UI	
AUD/2026/0000000328	10/05/2026 02:05:55	Failed Login	User	admin		0	Administrator	Web UI	

Failed operations

8. Alerts

Audit → Configuration → Alert Rules define when to warn; **Audit → Alerts** lists what happened. An alert is created in the same transaction as its evidence, so it can never describe a rolled-back change, and is delivered by a cron every 5 minutes as a chatter notification and a *To Do* activity for each recipient.

Audit Dashboard Audit Logs Alerts Reports Configuration						
New	Alert Rules		Search...		1-7 / 7	
☐	Name	Trigger	Model Name	Field Name	Severity	Active
☐	Journal entry deleted	Record deleted	account.move		High	☒
☐	Journal entry deletion blocked	Failed operation (e.g. blocked delete)	account.move		High	☒
☐	Invoice total changed > 10%	Field changed	account.move	amount_total	High	☒
☐	Bank account number changed	Field changed	res.partner.bank	acc_number	Critical	☒
☐	User groups changed	Field changed	res.users	groups_id	High	☒
☐	Mass deletion	Many deletions by one user			Critical	☒
☐	Brute force login	Repeated failed logins			Critical	☒

Default alert rules

Field	Meaning
Trigger	Record deleted · Failed operation · Field changed · Many deletions by one user · Repeated failed logins · Business action executed
Severity	Low, Medium, High, Critical.
Model Name · Field Name · Action Method	Technical names; empty model = every model.
Threshold type · Threshold	For <i>Field changed</i> : any change, absolute difference, or percentage (e.g. 10 %).
Count · Window (minutes)	For mass deletion and failed logins: e.g. 50 deletions in 10 minutes.
Deduplicate (minutes)	Same alert again within this time increases <i>Occurrence Count</i> instead of creating a new one (default 30).
Max alerts / hour	Above this, new alerts are stored as <i>Suppressed (rate limit)</i> (default 20). Prevents alert storms.
Recipients · Recipient Group	Who is notified. Empty = all Audit Managers.

Default alert rule	Trigger	Severity
Journal entry deleted	Record deleted, account.move	High
Journal entry deletion blocked	Failed operation, account.move	High
Invoice total changed > 10%	Field changed, account.move / amount_total, 10 %	High

Bank account number changed	Field changed, res.partner.bank / acc_number	Critical
User groups changed	Field changed, res.users / groups_id	High
Group members changed (inactive)	Field changed, res.groups / users	High
Mass deletion	50 deletions in 10 minutes	Critical
Brute force login	10 failed logins	Critical

Alert states: **Pending** → **Notified** → **Acknowledge** → **Acknowledged** → **Close** → **Closed** . The smart button **Evidence** opens the events behind the alert; write the investigation in **Note**.

Audit Dashboard Audit Logs Alerts Reports Configuration 4 13 My Company A					
Alerts Q Open Search... 1-13 / 13					
☐	Last Seen	Name	Rule	Severity	Occurren... State
☐	10/05/2026 14:18:39	Account Number changed on EG380019000500000000263180002: → EG380019000500000000263180002	Bank acco...	Critical	1 Notified
☐	10/05/2026 14:18:39	account.move Draft Invoice deleted by admin	Journal ent...	High	1 Notified
☐	10/05/2026 14:18:39	Total changed on Draft Invoice: → 1,552.50 USD	Invoice tot...	High	1 Notified
☐	10/05/2026 14:18:38	Failed unlink on account.move by admin: UserError: You can't delete a posted journal item. Don't play games with your accounting records; re...	Journal ent...	High	1 Notified
☐	10/05/2026 14:18:38	Total changed on Draft Invoice INV/2026/00003: 11,500.00 USD → 14,375.00 USD	Invoice tot...	High	1 Notified
☐	10/05/2026 14:18:38	Total changed on Draft Invoice: → 11,500.00 USD	Invoice tot...	High	1 Notified
☐	10/05/2026 14:17:49	Account Number changed on EG380019000500000000263189999: EG380019000500000000263180002 → EG38001900050000000026318...	Bank acco...	Critical	1 Notified
☐	10/05/2026 14:17:49	Account Number changed on EG380019000500000000263180002: → EG380019000500000000263180002	Bank acco...	Critical	1 Notified
☐	10/05/2026 14:16:55	10 failed logins for admin from 203.0.113.77	Brute force...	Critical	3 Notified
☐	10/05/2026 14:16:55	Groups on Hany Sales (demo) - removed: Administration / Settings	User group...	High	1 Notified
☐	10/05/2026 14:16:55	Groups on Hany Sales (demo) - added: Administration / Access Rights, Bypass HTML Field Sanitize, Administration / Settings, Canned Respo...	User group...	High	1 Notified
☐	10/05/2026 14:16:52	Groups on Rania Accountant (demo) - added: User types / Internal User, Extra Rights / Technical Features, Technical / Access to export featur...	User group...	High	1 Notified
☐	10/05/2026 14:16:52	Groups on Hany Sales (demo) - added: User types / Internal User, Extra Rights / Technical Features, Technical / Access to export feature, Extr...	User group...	High	1 Notified

Open alerts

Audit

Dashboard

Audit Logs

Alerts

Reports

Configuration

4

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My Company

A

Alerts

Total changed on Draft Invoice INV/2026/00003: 11,500.00 ...

Evidence

1 / 1

Acknowledge

Close

Pending

Notified

Acknowledged

Closed

Total changed on Draft Invoice INV/2026/00003: 11,500.00 USD → 14,375.00 USD

Rule

Invoice total changed > 10%

Severity

High

First Seen

10/05/2026 14:18:38

Last Seen

10/05/2026 14:18:38

Occurrence Count

1

Investigation notes

Reference	Event Datetime	Operation	Model	Record	Action	Fields	User	Source	
AUD/2026/0000000360	10/05/2026 14:18:38	Update	Journal Entry	Draft Invoice INV/2026/00003		12	Administrator	Web UI	

Send message

Log note

Activities

Planned Activities

A

Today: "Total changed on Draft Invoice INV/2026/00003: 11,500.00 USD → 14,375.00 USD" for Administrator

✓ Mark Done

✎ Edit

✕ Cancel

Today

An alert with its evidence: invoice total changed by 25 %

9. Sensitive fields and redaction

9.1 Sensitive field patterns

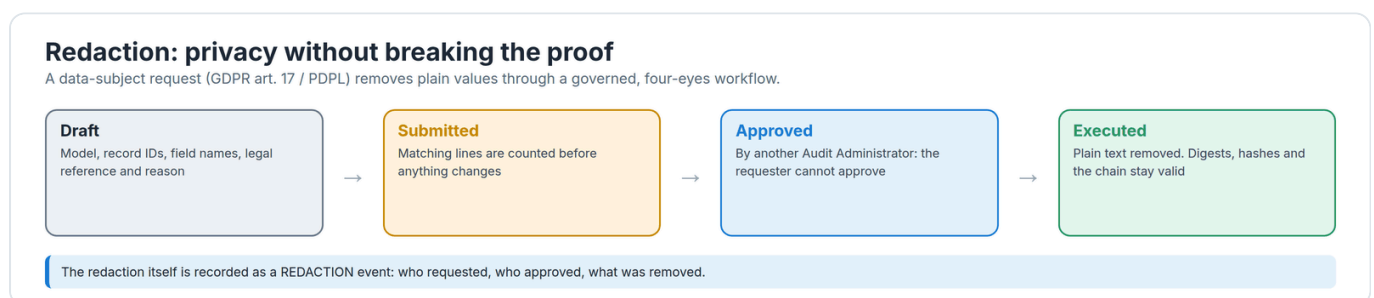
Audit → Configuration → Sensitive Fields: regular expressions on model and field names, applied to every rule.

- **Secret (never stored):** passwords, tokens, API keys, payment provider credentials. The event says the field changed; the value is masked. Built-in secret names (password, token, api_key, secret, otp...) are always masked even without a pattern.
- **Personal data:** emails, phones, bank account numbers, identification numbers, birth dates, postal addresses. Mode *Store value (redactable)* keeps the value and flags the line as personal; mode *Store keyed digest only* keeps only a digest.

Audit Dashboard Audit Logs Alerts Reports Configuration						
New Sensitive Fields Search... 1-9 / 9						
☐	Name	Category	Personal Mode	Model Pattern	Field Pattern	Active
☐	Passwords and secrets	Secret (never stored)			(^L)(password secret token api_?key private_?key otp totp)s?(\$L_)	☒
☐	API key value	Secret (never stored)		^res\.users\.apikey\$	^key\$	☒
☐	Payment provider credentials	Secret (never stored)		^payment\.provider\$	_(key secret password token)	☒
☐	Email addresses	Personal data	Store value (redactable)		(^L_)email(\$L_)	☒
☐	Phone numbers	Personal data	Store value (redactable)		(^L_)(phone mobile)(\$L_)	☒
☐	Bank account numbers	Personal data	Store value (redactable)	^res\.partner\.bank\$	^acc_number\$	☒
☐	Identification numbers	Personal data	Store value (redactable)		(^L_)(vat ssnid identification_id passport_id national_id)(\$L_)	☒
☐	Birth dates	Personal data	Store value (redactable)		(^L_)(birthday birthdate date_of_birth)(\$L_)	☒
☐	Postal addresses	Personal data	Store value (redactable)	^res\.partner\$	^(street street2 zip)\$	☒

Default sensitive field patterns

9.2 Redaction requests



Four-eyes redaction

- 1 **Audit → Configuration → Redaction Requests → New** : name, reason, legal reference, model, record IDs (empty = all), field names, optional date limit. The form shows how many lines match.
- 2 **Submit for approval** .
- 3 Another Audit Administrator clicks **Approve** (or **Reject**). The requester cannot approve their own request.

4

Execute redaction : plain values are removed; digests and hashes stay, so integrity verification still passes. A REDACTION event records the operation.

Audit

Dashboard

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13

My Company

A

New

Redaction Requests

[Demo] Erasure request - former contact

1 / 1

Submit for approval

Draft

Waiting Approval

Approved

Executed

[Demo] Erasure request - forme

SCOPE

Model Name ?

res.partner

Record IDs ?

0

Field Names ?

email,phone

Date To ?

Matched Line Count

0

Data subject request (GDPR art. 17 / PDPL). Demo only.

GOVERNANCE

Legal Reference ?

DSR-2026-0042

Requested By

Administrator

Approved By

Executed At

Redacted Line Count

0

Send message

Log note

1

Following

A

Administrator

Today at 2:16 PM

Audit Redaction Request created

A redaction request (demo)

10. Integrity: sealing and verification

Why evidence cannot be silently edited

Each sealed event carries the hash of the previous one. Changing, deleting or inserting a row breaks the chain at that exact point.

INTACT CHAIN

Event #1
previous hash: genesis
hash: 9f2c...

→

Event #2
previous hash: 9f2c...
hash: 41ab...

→

Event #3
previous hash: 41ab...
hash: c07e...

→

Event #4
previous hash: c07e...
hash: 77d1...

SOMEONE EDITS EVENT #3 IN SQL

Event #1
previous hash: genesis
hash: 9f2c...

→

Event #2
previous hash: 9f2c...
hash: 41ab...

→

Event #3
previous hash: 41ab...
hash: e5d0 ≠ c07e

→

Event #4
previous hash: c07e ≠ e5d0
hash: 77d1...

Its recomputed hash no longer matches → verification reports **hash mismatch** on #3 and a **broken link** on #4.

Key: nx_audit_hmac_key in odoo.conf. Values are hashed as keyed digests, so a governed redaction removes plain text without breaking the chain.

The hash chain

- Events are committed **unsealed**. Every 5 minutes the sealing cron links them, in sealing order, into one chain per company and year (c1-2026) or shared-2026 for records without company, using HMAC-SHA256 with the key from odoo.conf .
- Every day the verification cron recomputes each chain. You can also run it now: **Audit → Reports → Integrity Checks → New → Verify now** .
- Violation kinds: *Missing event, Broken link, Modified event, Modified value, Invalid checkpoint, Truncated tail, Unsealed backlog*.
- **Audit → Reports → Hash Chains** shows the last sequence and hash of each chain. For non-repudiation, export the chain heads regularly to external immutable storage.

Daily verification

Chain Count 4

Checked Count 383

Violation Count 0

Unsealed Backlog 0

Kind	Chain Key	Seal Seq	Reference	Detail

Hash Chains

Chain Key

Last Seq

Last Hash

Last Sealed At

c1-2026	40	78538a6479a1b53e9f755797578cc22864247a7859b96070440aa281e3a0dc1	10/05/2026 14:19:00
demo-c1-2026	17	6c5a385ea8929f403ce8540a50ee0ab3723ae12488a3944d7bc19a262cda4d5	10/05/2026 14:16:55
demo-shared-2026	316	c7499f9490c0f5b0b83eeb1e13cc063b340b1a8564f56a3d01ee4d737c54	10/05/2026 14:16:55
shared-2026	10	730e44933357789ad7d5d1cc3f5ee405917c061ed64199d34032f654ad44413	10/05/2026 14:19:00

A passed verification

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Audit

Dashboard

Audit Logs

Alerts

Reports

Configuration

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My Company

A

Audit Logs

AUD/2026/0000000352 UPDATE INV/2026/00003

Same request

1 / 1

Record timeline

Open record

Unsealed

Sealed

AUD/2026/0000000352

Update

INV/2026/00003

WHAT

Model

Journal Entry

Model

account.move

Record ID

4

Rule

Journal entries & invoices

Change Origin

Direct

WHO / WHEN

Event Datetime

10/05/2026 14:18:38

User

Administrator

User Login

admin

Initiating User ?

Administrator

Superuser Mode

☐

Company

My Company

Visible to Companies

My Company

Result

Success

Changes

Child events

Snapshot

Request

Integrity & retention

INTEGRITY

Chain Key

c1-2026

Seal Seq

10

Sealed At

10/05/2026 14:19:00

Hash

3da72da69875b65aa39e322f96523f0607e6c17e1949916865c284679c7c87b5

Previous Hash

5cebc3735f00c1f6e531bf83b977a4e6ba711f9f4aef9873a833fe2486e2e012

Hash Version

1

Hmac Key

1fb040084258

Snapshot Digest

e5e19ae2f4042fa844e0674e2f61c6be1fd3d8bdc1d17bbcc82ca9fcba2d55

RETENTION

Retention Policy

Finance (10 years, no purge)

Archive After

10/04/2028 14:18:38

Purge After

Prohibit Purge

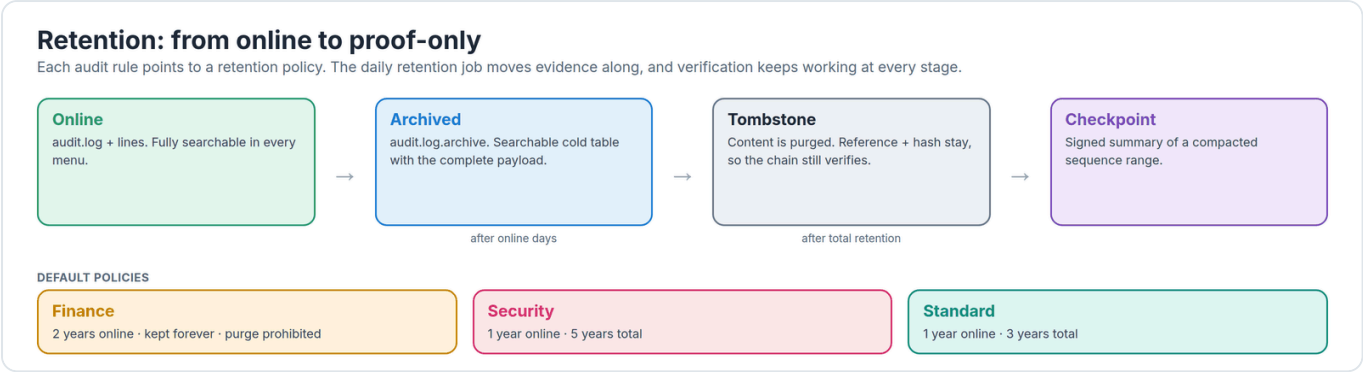
☒

Redacted

☐

Integrity & retention tab of a sealed event

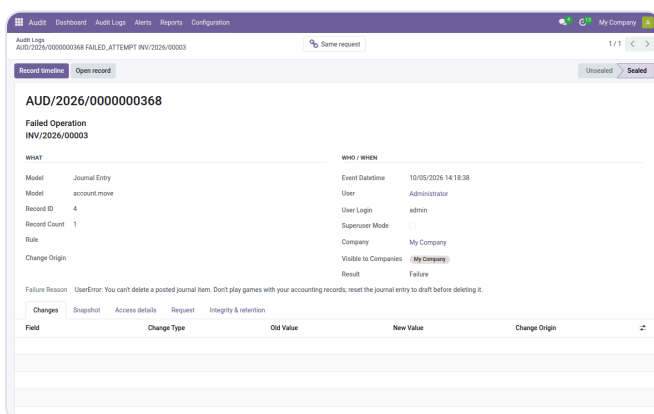
11. Retention policies



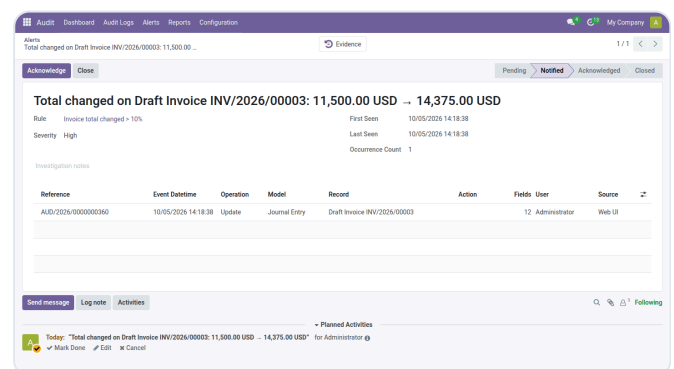
12. Hands-on example: catch a changed invoice

Goal: an accountant posts an invoice, resets it to draft, raises the price and posts it again; a colleague then tries to delete it. You will see the evidence, the alert, the timeline and the integrity proof. Do it on a test database (with Accounting installed).

- 1 Prepare.** In **Audit → Configuration → Audit Rules** check that *Journal entries & invoices* exists; if not, click **Load default rules** in the settings. In **Audit → Configuration → Alert Rules** open *Invoice total changed > 10%* and add yourself in **Recipients**.
- 2 Create the case.** **Invoicing → Customers → Invoices → New**: customer *Al Nour Trading*, 10 × product at 1 000. **Confirm**. Then **Reset to Draft**, change the price to 1 250, **Confirm** again.
- 3 See the evidence.** **Audit → Audit Logs → Updates**, filter on *Journal Entry*. Open the event: the **Changes** tab shows the price (direct) and the totals (derived) with old → new values, the user, IP and source *Web UI*.
- 4 See the alert.** Within 5 minutes you receive a *To Do* activity and a notification. **Audit → Alerts: Total changed on ... 11,500.00 → 14,375.00**. Click **Evidence**, then **Acknowledge** and write a note.
- 5 See the story.** On the invoice, **Action ⚙ → Audit Trail**: creation, posting, reset, price change and new posting, in order.
- 6 Try to hide it.** Try to delete the posted invoice: Odoo refuses. **Audit → Audit Logs → Failed Operations** shows the attempt with the error message, and the alert *Journal entry deletion blocked* is raised.
- 7 Prove it.** **Audit → Configuration → Settings → Seal pending events**, then **Audit → Reports → Integrity Checks → New → Verify now**: **Passed**.



Step 6: the blocked deletion



Step 4: the alert

Second exercise: give a user the *Administration / Settings* group, then remove it. You get two *User groups changed* alerts (High) and the change appears on both the user and the group.

13. For developers

```
from odoo.addons.nx_audit_log.services.api import audit_action, audit_bypass,
record_action

class SaleOrder(models.Model):
    _inherit = 'sale.order'

    @audit_action('Approve discount')          # semantic ACTION + linked field changes
    def action_approve_discount(self):
        ...

    def _sync_from_erp(self):
        with audit_bypass('ERP mirror sync - audited on the ERP side'):  # logged in
            server log
        ...

record_action(records, 'Recalculate commissions', method='_recalc') # imperative variant
```

- Override `_audit_dynamic_secret_fields(self, row)` on a model to mask values per row.
- Tests: `odoo-bin -d test_db -i nx_audit_log --test-enable --test-tags nx_audit --stop-after-init`
- Direct SQL writes bypass the ORM: `tools/sql_coverage_scan.py` lists them per module (see `doc/COVERAGE.md`).

14. Operations, limitations and FAQ

Scheduled action	Frequency	Purpose
Audit: seal events (hash chain)	5 minutes	Seal committed events
Audit: dispatch alerts	5 minutes (and on demand)	Notify recipients
Audit: retention (archive / purge / checkpoints)	Daily	Apply retention policies
Audit: verify integrity	Daily	Recompute all chains

Known limitations

- Raw SQL writes are not seen (e.g. reconciliation flags on journal items); the inventory is in `doc/COVERAGE.md`. Password changes are covered by a dedicated hook.
- Data loaded during module install / upgrade is not audited.
- Session expiry is not an event; only explicit logout is.
- Deleting a company or user referenced by sealed evidence is reported by verification; archive them instead.
- A PostgreSQL superuser can rewrite rows and hashes together: export chain heads to external storage for non-repudiation.
- Uninstalling the module drops the audit tables; it first exports every table to `<data_dir>/nx_audit_log_exports/<db>/<timestamp>/` (gzip JSON lines).

FAQ

Question	Answer
A new app is installed but its documents are not audited.	Click Load default rules , or create a rule for the model.
Too many events on a model.	Use <i>Selected fields only</i> , <i>Direct changes only</i> , a domain, or exclude technical users.
Dashboard shows events waiting for sealing.	Check that the sealing cron is active, or click Seal pending events .
The settings warn about the integrity key.	Add <code>nx_audit_hmac_key</code> to <code>odoo.conf</code> and restart.
Search in old/new values is slow.	Enable value search indexes (needs <code>pg_trgm</code>).
Can someone delete audit logs?	No. No group has create/write/delete rights on evidence, and the models refuse writes outside internal operations (seal, archive, purge, redact).